

Summary

Company Name & Members: LFAssociates-Abigail Marciniak, Brian Pham Ngo, Daniyal Rauf, Imart De Guzman, Isaac Min Bae, Shreya Thota

Potential device name: Taenacity

Disparity: Lack of healthcare accessibility due to **poverty** and **neglected diseases in tropical regions**.

- People living in tropical regions worldwide are prone to developing various environment-related illnesses, many of which do not receive proper funding by the current healthcare system. As a result, accurate and easily accessible POC devices have not yet been developed, leading to a significant number of casualties in impoverished tropical areas.

Problematic disease: **T. Solium Taeniasis and Cysticercosis**

- Approximately 2.5 million people are infected with T. Solium (tapeworm & larvae) ([Taenia solium Taeniasis and Cysticercosis in Southeast Asia](#))
- Neurocysticercosis—caused by the ingestion of T. Solium tapeworms—accounts for 50,000 deaths annually and is the leading cause of epilepsy (WHO).
- T. Solium Taeniasis typically asymptomatic, but can cause headache, nausea, abdominal pain & discomfort.
- Incidence rates vary by region, however it is most common in the tropic regions of Asia, Africa, and Latin America. (E.g a study in West Africa recorded prevalence rates ranging from 0.68% to 21.63% for human cysticercosis, while taeniasis had ratings from 0.3% to 13.7%)([Prevalence and factors associated with human Taenia solium taeniosis and cysticercosis in twelve remote villages of Ranomafana rainforest, Madagascar](#))

Current Detection Strategies:

- One T. Solium point-of-care prototype is a three-step in-house-produced rapid diagnostic test (RDT) for the simultaneous detection of taeniosis (TST) and cysticercosis (TSCC) antibodies in a lateral-flow format based on two recombinant proteins, rES33 and rT24H, respectively.
 - Although the preliminary results had high sensitivity and specificity values (TST: Sensitivity: 88-93%, Specificity: 99%. TSCC: Sensitivity: 82%, Specificity: 99%),
- There is no current point of care device for either Taeniasis or Cysticercosis on the market.
- Traditional detection strategies:
 - Stool microscopy: Microscopic demonstration of ova and proglottids, usually inside a fecal sample. To improve test sensitivity, three fecal samples collected on different days should be analyzed. Since T. Solium needs subsequent testing, stool microscopy that allows for species identification is recommended.
 - Serologic Diagnosis: Serology is an enzyme-linked immunoelectrotransfer blot (EITB) test which can be performed using serum or cerebrospinal fluid (CSF). Serologic testing varies in sensitivity for infections so a negative result should not be left alone.
 - Other strategies including coproantigen, nucleic acid-based, and serologic testing are only available on a research basis.

Product: **Lateral flow test** using gold nanoshells to increase sensitivity, giving more accurate results. (Design further explained in schematic below)

Estimated Costs:

- Current tests can be as expensive as \$503 for neurocysticercosis, however studies show that lateral flow tests can cost as low as \$0.10 - \$3.00 per test.
- The development costs for lateral-flow immunoassays are also low and range from \$30 K to \$100 K maximum to invest and develop a target analyte in this case for T. Solium diseases.

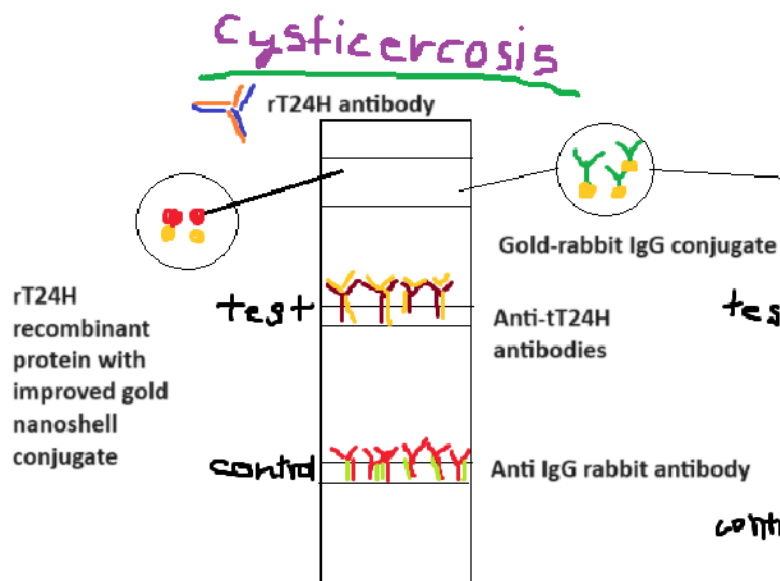
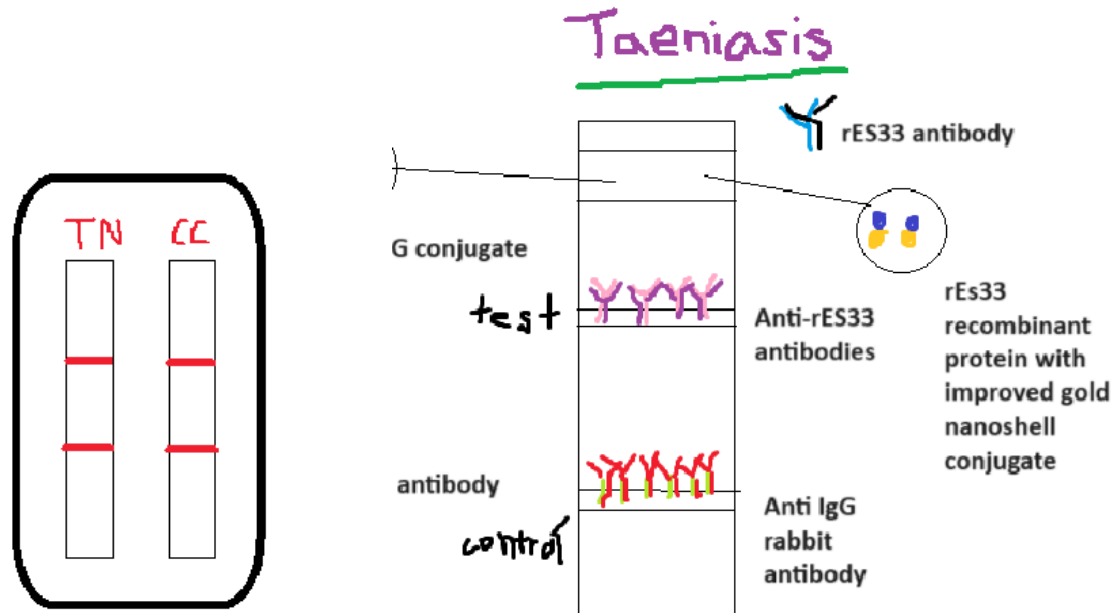
Innovativeness of Device:

- **Point-of-care capacity:** Unlike traditional detection methods that require laboratory facilities and trained personnel, our device is designed as a point-of-care device. This means it can be used directly at the patient's location, enabling rapid diagnosis and immediate initiation of treatment, especially in remote and underserved tropical regions where access to healthcare infrastructure is limited.
- **Gold nanoshells:** The incorporation of gold nanoshells in the lateral flow test isn't seen in products or devices on the market that address the same issue. Gold nanoparticles offer unique optical properties, including high sensitivity to molecular binding events, which can enhance the accuracy and sensitivity of the diagnosis test. This innovative use of nanotechnology contributes to the device's effectiveness in detecting T. Solium Taeniasis and Cysticercosis.
- **Increased sensitivity:** By utilizing gold nanoshells, our device achieves increased sensitivity compared to traditional detection methods. This heightened sensitivity allows for the detection of lower levels of antibodies associated with T. Solium Taeniasis and Cysticercosis, improving the reliability of diagnosis and reducing the likelihood of false-negative results.
- **Cost-effectiveness:** Our device aims to address the affordability barrier associated with existing diagnostic tests. While current tests for neurocysticercosis can be expensive, our device offers a cost-effective alternative, with estimated costs ranging from \$0.10 to \$3.00 per test. This affordability makes the device accessible to healthcare providers and patients in resource-limited settings, where the burden of these diseases is often the greatest.

Regulatory Plans, Testing, etc...

- ISO 13485 is the international set standard of quality management for any medical device, and their requirements must be followed by any company working in manufacturing these products.
- Protocol requires specific physical-chemical characterization (morphology, shape, surface change), in vitro characterization (i.e., sterility, drug release, targeting, toxicity, etc.), and in vivo characterization (i.e., efficacy, exposure, etc.).
- Limitations: Cross reactivity, Once sample is applied the action cannot be sped up
- Ensure that clinical trials follow the GCP guidelines
- Conduct FTO analysis to assess if the lateral flow test infringes on other current patents
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- 510(k) Premarket Notification:
 - 510(k) Number: K031942
- Testing plan:
 - Prototype testing: See if there are two distinct lines on the device
 - Device efficacy (Sensitivity and Specificity):
 - Sample size: ~20 End users, ~ 1000 tests with device
 - Setting: If budget allows: Eastern Europe, Russia, eastern Africa, or Latin America. If not: States in United States where free roaming pigs and people are concentrated and sanitation is poor

Design Schematic



We propose a multiplexed lateral flow assay device capable of detecting the presence of rES33 antibodies (present in Taeniasis patients) and rT24H antibodies (associated with cysticercosis). The sample pads contain a gold-rabbit IgG conjugate and gold nanoshells attached to the rT24H and rES33 proteins. The gold rabbit IgG conjugate antibodies will travel down the assay through capillary action and bind to the Anti-rabbit IgG antibodies on the control line. If there are antibodies present in the sample, they will create a complex with the nanoshell + protein and bind to the “anti-antibodies” on the test line. By using 150 nm gold nanoshells as our optical indicator as opposed to the 40 nm gold nanoparticles (industry standard), we can dramatically increase the sensitivity of the device (as opposed to the existing prototype).

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